

Paper 1 - Set 06 - Solutions

Generated practice paper: answer key and worked solutions.

Answer key

Question	Answer	Topic
1	C	Quadratic functions
2	C	Integration
3	A	GCSE algebra
4	B	Indices and surds
5	D	Differentiation
6	C	Straight lines
7	B	Circles and circle theorems
8	E	Differentiation
9	D	Sequences and series
10	C	Probability
11	B	Binomial expansion
12	D	Circles and circle theorems
13	A	Functions and transformations
14	F	Exponentials and logarithms
15	C	Angles, polygons, congruence and vectors
16	E	Integration
17	B	Polynomials, factor and remainder theorems
18	E	Angles, polygons, congruence and vectors
19	C	Triangles, radians and sectors
20	A	Indices and surds

Worked solutions

Question 1 QUA-05 · Quadratic functions · spec MM1.3

For what value of c does the line $y = 2x + c$ touch the parabola $y = x^2 - 4x + 7$?

Answer: C

Equate: $x^2 - 6x + (7 - c) = 0$. Tangency needs discriminant zero: $36 - 4(7 - c) = 0$, so $8 + 4c = 0$ and $c = -2$.

Question 2 INT-04 · Integration · spec MM7.6

A curve has $\frac{dy}{dx} = 6x^2 - 4x + 1$ and passes through $(1, 3)$. Find its equation.

Answer: C

$y = 2x^3 - 2x^2 + x + c$; at $(1, 3)$: $2 - 2 + 1 + c = 3 \Rightarrow c = 2$.

Question 3 ALG-04 · GCSE algebra · spec M4.8

Which of the following is an identity (true for all values of x)?

Answer: A

$(x + 3)^2 - (x - 3)^2 = (x^2 + 6x + 9) - (x^2 - 6x + 9) = 12x$ for all x .

Question 4 IND-19 · Indices and surds · spec MM1.1

How many real solutions does the equation $2^{x^2} = 16 \times 2^{-3x}$ have?

Answer: B

$2^{x^2} = 2^{4-3x}$, so $x^2 + 3x - 4 = 0$, i.e. $(x + 4)(x - 1) = 0$: two real solutions.

Question 5 DIF-01 · Differentiation · spec MM6.2

Find $\frac{dy}{dx}$ when $y = \frac{(3x - 2)^2}{x^2}$.

Answer: D

$y = 9 - 12x^{-1} + 4x^{-2}$, so $y' = 12x^{-2} - 8x^{-3}$.

Question 6 LIN-01 · Straight lines · spec MM3.1

Find the equation of the line through $(2, -3)$ that is perpendicular to $3x - 2y + 5 = 0$.

Answer: C

The given line has gradient $\frac{3}{2}$, so the perpendicular has gradient $-\frac{2}{3}$:
 $y + 3 = -\frac{2}{3}(x - 2) \Rightarrow 3y + 9 = -2x + 4 \Rightarrow 2x + 3y + 5 = 0$.

Question 7 CIR-01 · Circles and circle theorems · spec MM3.2

Find the radius of the circle $x^2 + y^2 - 6x + 4y - 12 = 0$.

Answer: B

$(x - 3)^2 + (y + 2)^2 = 12 + 9 + 4 = 25$, so $r = 5$.

Question 8 DIF-20 · Differentiation · spec MM6.3

Find the sum of the y -coordinates of the stationary points of the curve $y = 2x^3 - 9x^2 + 12x - 3$.

Answer: E

$y' = 6x^2 - 18x + 12 = 6(x-1)(x-2)$: stationary points at $x = 1$ ($y = 2 - 9 + 12 - 3 = 2$) and $x = 2$ ($y = 16 - 36 + 24 - 3 = 1$). Sum 3.

Question 9 SEQ-03 · Sequences and series · spec MM2.1

A sequence is defined by $x_{n+1} = 3x_n - 2$ with $x_1 = 2$. Find x_5 .

Answer: D

$x_2 = 4, x_3 = 10, x_4 = 28, x_5 = 82$.

Question 10 PRO-02 · Probability · spec M7.6

Two fair dice are thrown. Find the probability that the product of the two scores is odd.

Answer: C

The product is odd only if both scores are odd: $\frac{1}{2} \times \frac{1}{2} = \frac{1}{4}$.

Question 11 BIN-10 · Binomial expansion · spec MM2.4

Find the coefficient of x^5 in the expansion of $(1+x)^3(1+x)^4$.

Answer: B

$(1+x)^3(1+x)^4 = (1+x)^7$, and $\binom{7}{5} = 21$.

Question 12 CIR-06 · Circles and circle theorems · spec MM3.3

A, B and C lie on a circle with centre O . Angle $AOB = 100^\circ$ and C lies on the major arc AB . Find angle ACB .

Answer: D

The angle at the centre is twice the angle at the circumference subtended by the same arc: $ACB = 50^\circ$.

Question 13 FUN-01 · Functions and transformations · spec MM8.2

The graph of $y = x^2$ is translated by 3 units in the positive x -direction and then stretched parallel to the y -axis with scale factor 2. What is the equation of the resulting graph?

Answer: A

Translation: $y = (x-3)^2$. Stretch by factor 2 in the y -direction: $y = 2(x-3)^2$.

Question 14 LOG-03 · Exponentials and logarithms · spec MM5.3

Solve $2^{2x} - 3 \cdot 2^{x+2} + 32 = 0$. Find the sum of the solutions.

Answer: F

Let $y = 2^x$: $y^2 - 12y + 32 = (y-4)(y-8) = 0$, so $2^x = 4$ or 8 : $x = 2$ or 3 . Sum 5.

Question 15 GEO-15 · Angles, polygons, congruence and vectors · spec M5.2

A regular hexagon and a regular octagon share a common side. What is the size of the angle between their two other sides that meet at one end of the common side (the angle outside both polygons)?

Answer: C

Angles at the shared vertex: 120° (hexagon) $+135^\circ$ (octagon) $+x = 360^\circ$, so $x = 105^\circ$.

Question 16 INT-19 · Integration · spec MM7.1

The curve $y = x(x-1)(x-2)$ and the x -axis enclose two regions. What is the ratio of the area of the region for $0 \leq x \leq 1$ to the area for $1 \leq x \leq 2$?

Answer: E

The cubic has rotational symmetry about $(1, 0)$ (it is $u^3 - u$ with $u = x - 1$, an odd function), so the two regions are congruent: ratio 1 : 1.

Question 17 POL-09 · Polynomials, factor and remainder theorems · spec MM1.6

The expression $x^4 + 4$ can be factorised into two quadratics with integer coefficients. What is the sum of the coefficients of x in these two quadratics?

Answer: B

$x^4 + 4 = (x^2 + 2)^2 - 4x^2 = (x^2 + 2x + 2)(x^2 - 2x + 2)$. The x -coefficients are 2 and -2 , summing to 0.

Question 18 GEO-11 · Angles, polygons, congruence and vectors · spec M5.19

$\vec{AB} = 3\mathbf{p} + 2\mathbf{q}$ and $\vec{BC} = k\mathbf{p} - 4\mathbf{q}$, where $\mathbf{p}$ and $\mathbf{q}$ are not parallel. Given that A , B and C are collinear, find k .

Answer: E

Collinear means $\vec{BC} = \lambda\vec{AB}$: $-4 = 2\lambda \Rightarrow \lambda = -2$, so $k = 3\lambda = -6$.

Question 19 TRI-07 · Triangles, radians and sectors · spec MM4.1

In triangle ABC , $AB = 6$, $BC = 4$ and angle $BAC = 30^\circ$. How many possible triangles are there?

Answer: C

Sine rule: $\frac{\sin C}{6} = \frac{\sin 30^\circ}{4}$, so $\sin C = \frac{3}{4}$. Since $BC = 4 < AB = 6$ and $4 > 6\sin 30^\circ = 3$, both the acute and obtuse values of C give valid triangles: the ambiguous case, two triangles.

Question 20 IND-22 · Indices and surds · spec MM1.2

Which one of the following is the largest?

Answer: A

Each option is $\sqrt{a} + \sqrt{b}$ with $a + b = 9$. Squaring gives $9 + 2\sqrt{ab}$, which is largest when the product ab is largest, i.e. when a and b are as close as possible. Option E has $a = b = 4.5$ (product 20.25), beating $4 \times 5 = 20$ for option C. Numerically: $E = \sqrt{18} \approx 4.243$, $C = 2 + \sqrt{5} \approx 4.236$.